MANDAR JOSHI

Phone: +91 70202 12202 | **Email:** mandarjoshi1045@gmail.com
LinkedIn: linkedin.com/in/mandar-joshi-0b951b28a | **GitHub:** github.com/mandar1045
Portfolio: mandarjoshi.me

PROFESSIONAL SUMMARY

Software engineer with a deep passion for open-source development and scalable systems. Active contributor to major open-source projects including **Supabase, FOSSology, and Elixir**, focusing on core API reliability, secure self-hosted infrastructure, and distributed systems. Experienced in architecting production-grade platforms with Go, Elixir, C, PHP, Docker, and TypeScript, and always looking to solve complex engineering challenges in the open.

EDUCATION

Vellore Institute of Technology — Vellore, India
B.Tech in Information Technology — *Jul. 2024 – May 2028 (Expected)*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, Cloud Computing

TECHNICAL SKILLS

Languages: Go, Elixir, TypeScript, C, PHP, SQL, JavaScript, Bash
Frameworks: Phoenix, Node.js, Express.js, Next.js, React.js, React Native, gRPC/Protobuf, Tailwind CSS
Infra/Tools: Docker, Docker Compose, Kubernetes, Linux, Nginx, Caddy, Cloudflare, GitHub Actions, CI/CD
Databases/Other: PostgreSQL, Supabase, Redis, Apache Kafka, ExUnit, PHPUnit, Git, Postman

OPEN SOURCE

SUPABASE INC. — *13 Merged PRs, 14 Open PRs, 25 Issues*

- Secured self-hosted Realtime instances by moving hardcoded encryption keys to configurable `.env` variables, and added strict JWT audience/issuer validation in the Supabase Server.
- Proxied OAuth metadata endpoints (`.well-known` routes) to the API gateway in Caddy/Nginx configurations, resolving critical 404s for self-hosted deployments.
- Fixed cascading Docker Compose healthcheck failures by introducing proper grace intervals, and resolved multiple test flakes and log suppression bugs in Elixir-based Realtime backend.

FOSSOLOGY (LINUX FOUNDATION) — *11 Merged PRs, 3 Open PRs, 8 Issues*

- Eliminated N+1 queries, redundant DB lookups, and null-crashes in the `/scanned` API endpoint. Updated member API endpoints to adhere to proper REST conventions.
- Fixed severe SQL formatting bugs and type misuses in the DAO layer, backing fixes with comprehensive PHPUnit tests.
- Resolved critical memory bugs in the C-based `pkgagent`, including unsafe `strcpy()` usage, NULL-before-free crashes, and string-length tracking bugs.

ELIXIR (ELIXIR-LANG) — *3 Merged PRs, 4 Issues*

- Made `mix format` output stable for comments after operators and inside tuple arguments, with regression tests.
- Fixed the formatter dropping parens on anonymous calls with `do-end` blocks, which emitted unparseable code.

WORK EXPERIENCE

SOFTWARE DEVELOPER INTERN | STEALTH STARTUP — *Aug. 2026 – Present*

- Building production web and mobile applications for large-scale egocentric video datasets that power the next generation of robotics and embodied AI systems.
- Developing full-stack features and infrastructure across Next.js, React Native, Supabase, and Cloudflare.
- Building and supporting robust data pipelines and quality-assurance tooling to streamline high-throughput video dataset workflows for field teams and clients.

SOFTWARE ENGINEER INTERN | STEALTH STARTUP — *Dec. 2025 – Jun. 2026*

- Architected secure, high-performance RESTful APIs and microservices in Node.js and Express for enterprise applications.
- Built production-facing platform features across backend services, workflow automation, and AI-assisted product surfaces.
- Improved system scalability with queue-driven workflows and tighter coordination across product and engineering teams.

PROJECTS

Resync: UPI Autopay Recovery Platform | LINK — *2026*

- Architected a **9-microservice** payment recovery platform in **Go + gRPC** with smart dunning engine that classifies UPI failure codes and schedules context-aware retries.
- Orchestrated event-driven pipeline using **Kafka** (partition-keyed ordering), **Redis** distributed locks, and idempotency keys ensuring **zero double-charges**.

Continum: AI/ML Workflow Automation Platform | LINK — *2026*

- Deployed a containerized microservices architecture using Docker and Kubernetes for high availability.
- Established robust CI/CD pipelines for automated testing, building, and deployment, ensuring reliable releases.
- Automated Gmail drafting with no-code AI workflows that write context-aware replies and schedule meetings.